MILITARY SPECIFICATION

INTEGRATED CIRCUIT, DIGITAL, CMOS CONTROL AND TIMING BASE, MONOLITHIC, SILICON

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, test, manufacturing and acceptance requirements for the integrated circuit (I.C.) to be used in the M718E1/M741E1 Mines.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extend specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38510	-	Microcircuits, General Specification for.
MIL-I-45208	-	Inspection System Requirements.
MIL-M-55565	-	Microcircuits, Preparation for Delivery of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-883	-	Test Methods and Procedures for Microelectronics.
MIL-STD-1331	-	Parameters to be Controlled for the Specification of Micro-circuits.
MIL-STD-130	-	Identification Marking of U.S. Military Property.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extend herein.

DRAWING

U. S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

PRODUCT AND PACKAGING DRAWINGS

9342575	-	Logic Diagram, Integrated Circuit, Timing Base, CMOS.
9317646	-	Qualification, Electronic Component.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Parts and Materials.

3.1.1 Components. The I.C. shall comply with all the requirements of the applicable drawings, specifications and standards.

3.1.2 Metals. External metal surface shall be corrosion resistant or shall be plated or treated to resist corrosion. External leads shall meet the requirements specified in 3.5.3.

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3.1.3 Other materials. External parts, elements or coatings including markings shall not blister, crack, soften, flow or exhibit defect that adversely affect storage, operation or environmental capabilities of the integrated circuit delivered to this specification under the specified test condition.

3.2 Design and construction. I.C. design and construction shall be in accordance with the requirements specified herein and in the applicable specifications or drawings.

3.2.1 Dimensions. The package dimension shall conform to MIL-M-38510, Appendix C, Case Outline D6, Configuration 1.

3.2.2 Package. The finished component shall be supplied in a 18 pin, dual-in-line hermetically sealed package. The package shall be of the type shown in figure 16, utilizing a ceramic lid, a ceramic base with gold deposit for mounting the die, Alloy 42 or Kovar solid metal lead frame thru to cavity with aluminum cladding, and a glass frit to accomplish the seal between the lid, lead frame and the base frit seal temperature shall be in excess of 400°C. No organic or polymeric material shall be used inside the package.

3.2.3 Die configuration. Active circuit area shall be the area delineated as inside a 2 MIL border and shall not include the area within 1.5 MILS of the bonding pads.

3.2.4 Passivation. The chip shall be passivated with silicon dioxide and/or silicon nitrade to render it impervious to contaminated environments. The bonding pad areas shall be the only areas with no surface passivation.

3.2.5 Static charge protection. Static charge protection shall be provided internally on all terminals.

3.3 Design documentation. When specified in the procurement document, design, topography and schematic circuit information for the microcircuit supplied under this specification shall be submitted to the procuring activity. Unless otherwise specified, all design documentation shall be sufficient to depict completely the physical and electrical construction of the microcircuit supplied under this specification, and shall be traceable to the specific production lot(s) and inspection lot codes under which microcircuit are manufactured and tested so that revisions can be identified.

3.3.1 Die topography. For microcircuit dies, there shall be an enlarged color photograph(s) showing the topography of elements of the die to a minimum magnification of 80X. If this results in a photograph larger than 8" x 10", the magnification may be reduced to accommodate an 8" x 10" view.

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3.3.2 Logic diagram. The actual logic diagrams of the microcircuit supplied under this specification shall be provided, showing all the logic/circuit elements functionally designed into the microcircuit together with their values, when applicable.

3.4 Electrical requirements.

3.4.1 Maximum Rating. (As specified in Table 1)

Characteristic	Symbol	Rating	Unit
D C Supply Voltage (See 3.4.4)	V_B	5 to +10	Volts
All Inputs	V_{IN}	-0.5 to +10	Volts
Temperature			
Operating	$T_A(OP)$	-55 to +125°C	
Storage	$T_A(STG)$	-65 to +150°C	

TABLE 1 MAXIMUM RATING

3.4.2 Electrical characteristics. (As specified in Table 2)

TABLE 2 PERFORMANCE CHARACTERISTICS

Characteristic	Conditions	Min	Max	Unit
Capacitance coupled feedback signal amplitude Terminal 10	V_B=5.0V with terminal driven by a simulated TTB signal source (See 4.5.12)	0.5		V
Leakage Current	T_A = +25°C V_B = 7.5V		5.0	UA
	See 4.5.2 -32°C ≤ T_A ≤ +63°D V_B = 7.5V		32.0	UA
Dynamic Load Current (Average Current)	T_A = +25°C V_B = 7.5V		95.0	UA
	See 4.5.3 -32°C ≤ T_A ≤ +63°C V_B = 7.5V		120.0	UA
"1" Output Voltage: Terminal 15	Terminal 15 Connected to Ground Thru 1 KOHM See Figure 7 — V_B = 5.0V, T_A = +25°C	2.5		V
	V_B = 5.0V, T_A = +63°C	2.5		V

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Characteristic	Conditions	Min	Max	Units
"0" Output Voltage: Terminal 15	Terminal 15 Connected To V_B Thru 1 KOHM See Figure 8 V_B = 5.0V T_A = +25°C		2.5	V
	V_B = 5.0V T_A = +63°C		2.5	V
Sequential Operation	(Greater of LVD or 2.5 Volts) _ V_B _ 7.5 Volts) -32° _ T_A _ +63°C	See 3.4.9		
LVD Function Voltage	T_A = +25°C See 4.5.6	1.7	3.2	V
	-32°C T_A +63°C	1.5	3.5	V
Input Switching Voltage: Terminal 10	See 4.5.7 and 3.4.8	0.7	5.4	V
Input Resistance	Terminals 1, 5, 7 Connected to V_B thru 500K Ohms	2.0		V
	Terminal 2, Connected to Ground thru 500K Ohms See 4.5.8		3.8	V
"1" Output Voltage: Terminal 11	V_B = 5.0 V with Terminal Connected to Ground Through 50K Ohms See 4.5.9	2.5		V
"0" Output Voltage:	V_B = 5.0V with Terminal Connected to V_B Through 50K Ohms See 4.5.10		2.5	V

3.4.2.1 The I.C. circuit shall function in accordance with the logic diagram as specified in Dwg. 9342575 when operated under power supply requirements as specified in paragraph 3.4.4 and input signals as specified in paragraph 3.4.3. All resistors are ± 1% metal film, all capacitors are ± 10% Tantalum and all voltages are ± .05 volts.

3.4.3 Test pattern. The circuit shall produce the outputs of Figure 1A when exercised in accordance with the input pattern therein. This test to be conducted at 1 KHz or greater clock rate.

3.4.4 Power supply requirements. The component shall be capable of functioning with a power supply having the following requirements.

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3.4.4.1 The output impedance of the supply source shall be less than 10,000 ohms and shall be a 1.0 uf minimum capacitor across its output.

3.4.4.2 Noise voltage on the supply output shall not exceed 5% V_B peak to peak and its frequency shall not exceed 1 KH_Z.

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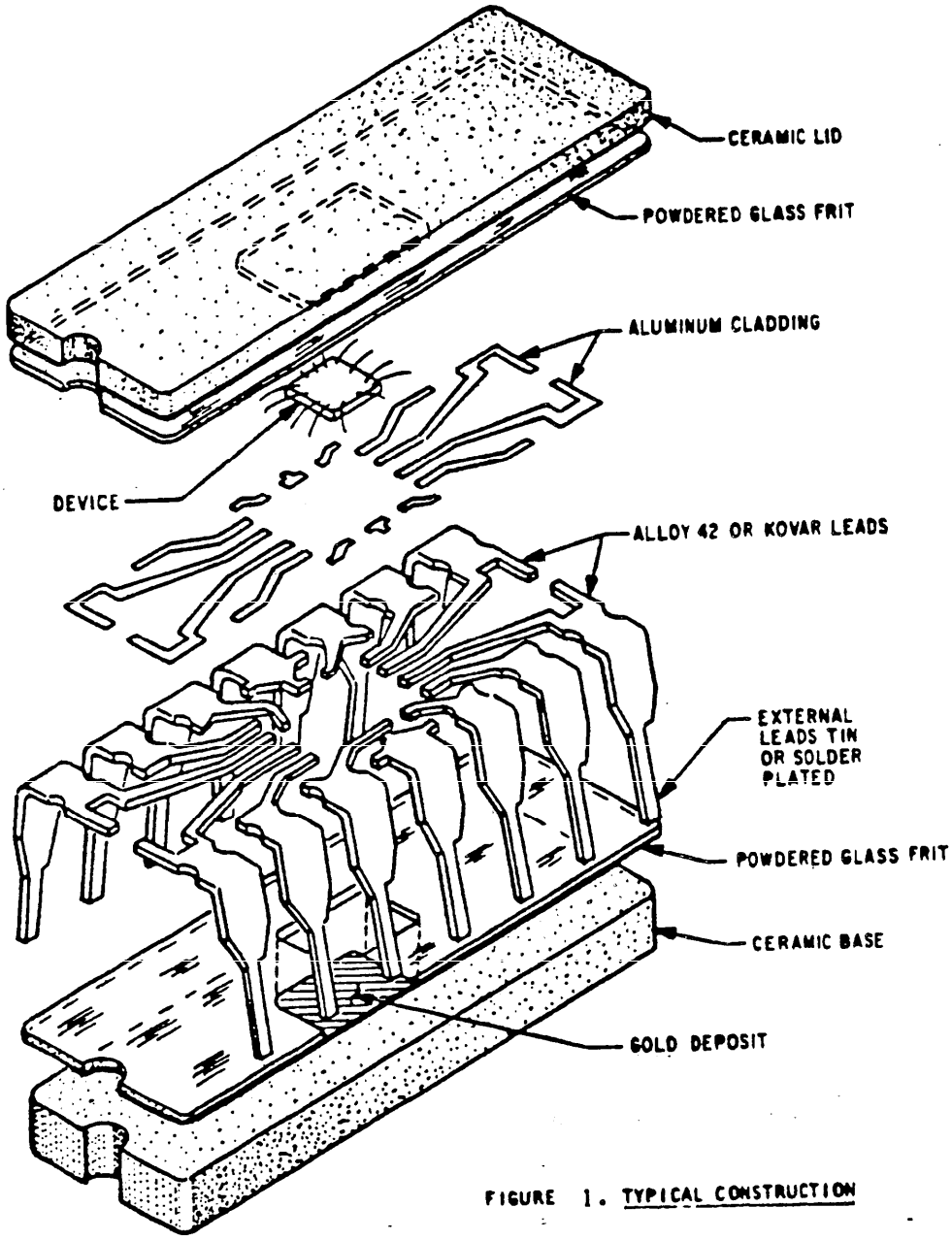


FIGURE 1. TYPICAL CONSTRUCTION

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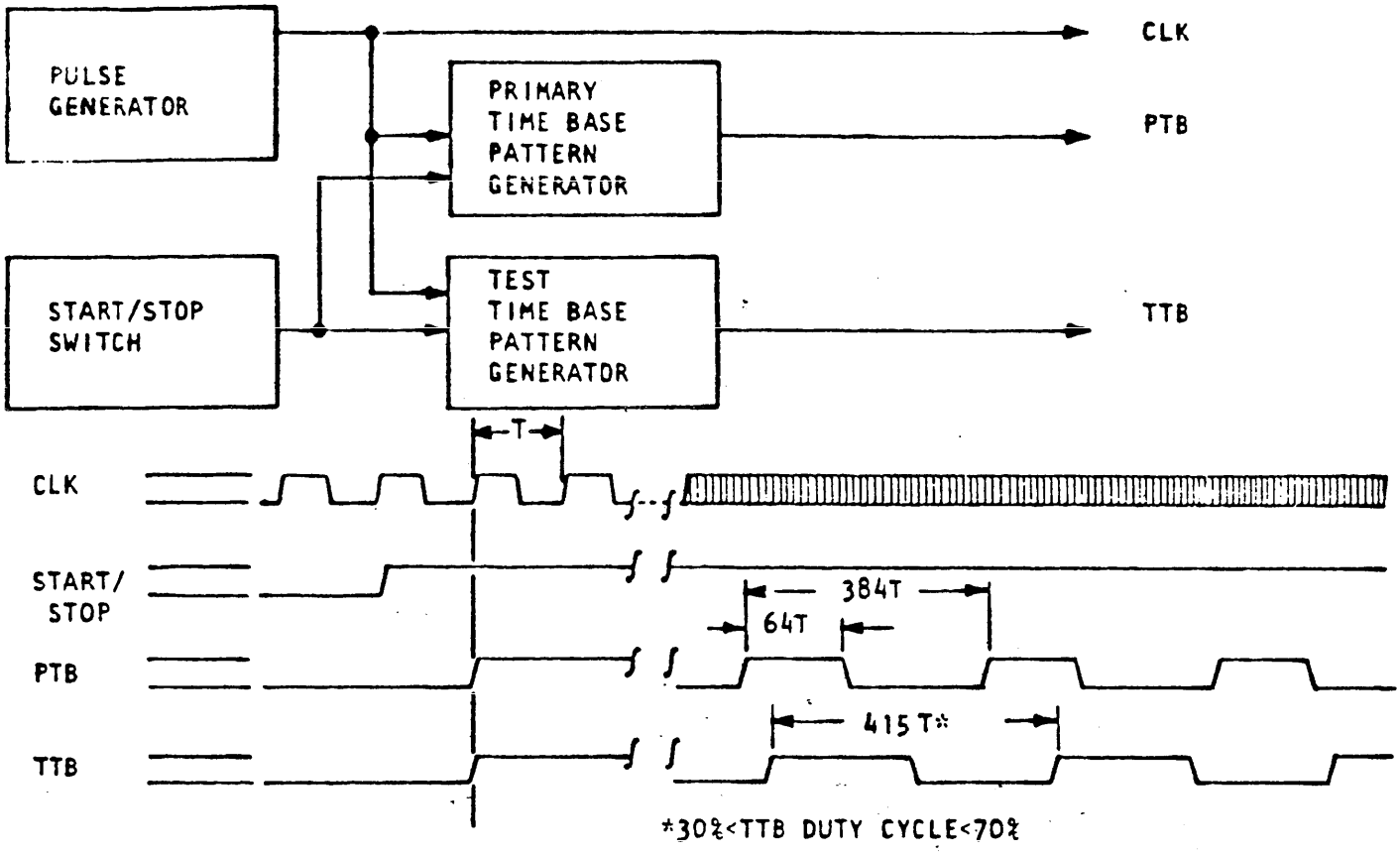


FIGURE 1A. INPUT PULSE PATTERN GENERATOR

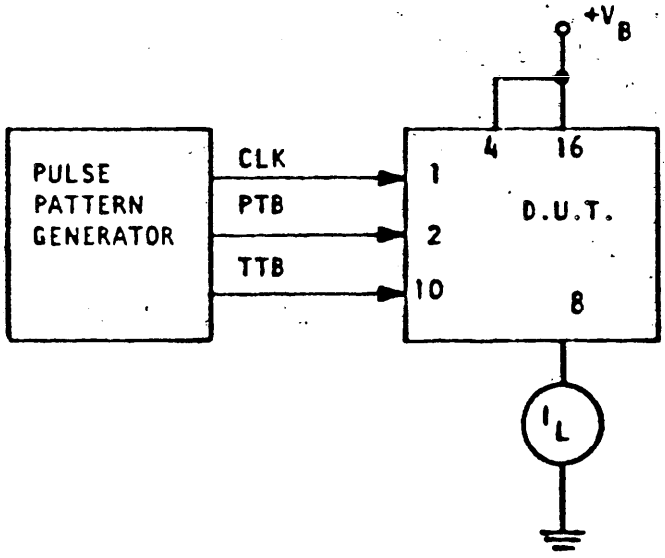


FIGURE 1B. DYNAMIC LOAD CURRENT TEST

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3.4.5 Input waveform. Input waveform shall have the following characteristic unless otherwise specified:

Rise Time (t_r1) =	5.0	Microsecond Max
Fall Time (t_f1) =	5.0	Microsecond Max
Frequency (f) =	10.0	KH_Z Max

3.4.6 Output wave form. Output signals produced by the component shall have the following characteristic unless otherwise specified:

Rise Time (t_r) =	5.0	Microsecond Max
Fall Time (t_f) =	5.0	Microsecond Max

The rise and fall of the output waveform shall be a smooth transition with no irregularities between 20% and 80% of full amplitude.

3.4.7 Electrical failure. Materials, processing, and layout design of the component shall be such that intermittency in the reset circuit and in the (2) ^{11} scaler is minimized, regardless of the failure mode.

3.4.8 The input switching voltage of Pin 10 shall be measured at room temperature by observing the output waveform on Pin 11.

3.4.9 Sequential operation.

3.4.9.1 When the component is connected as shown in Figure 2 and the waveforms shown in Figure 3 are applied to Terminals 1, 2, 4, 5, 7, 26, 10 and 14, operation shall be as described by the output waveforms in Figure 3.

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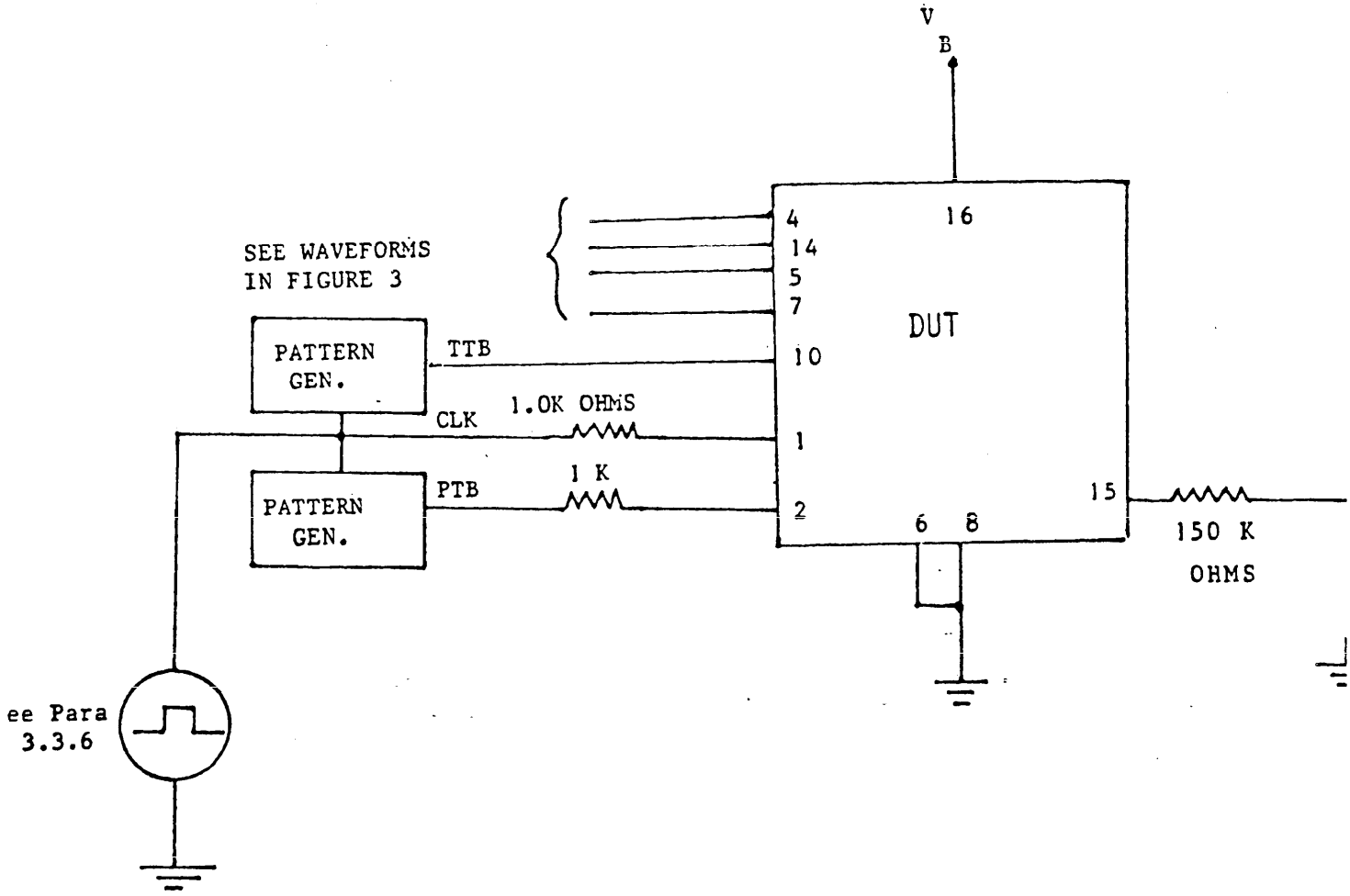


FIGURE 2 SEQUENTIAL OPERATION

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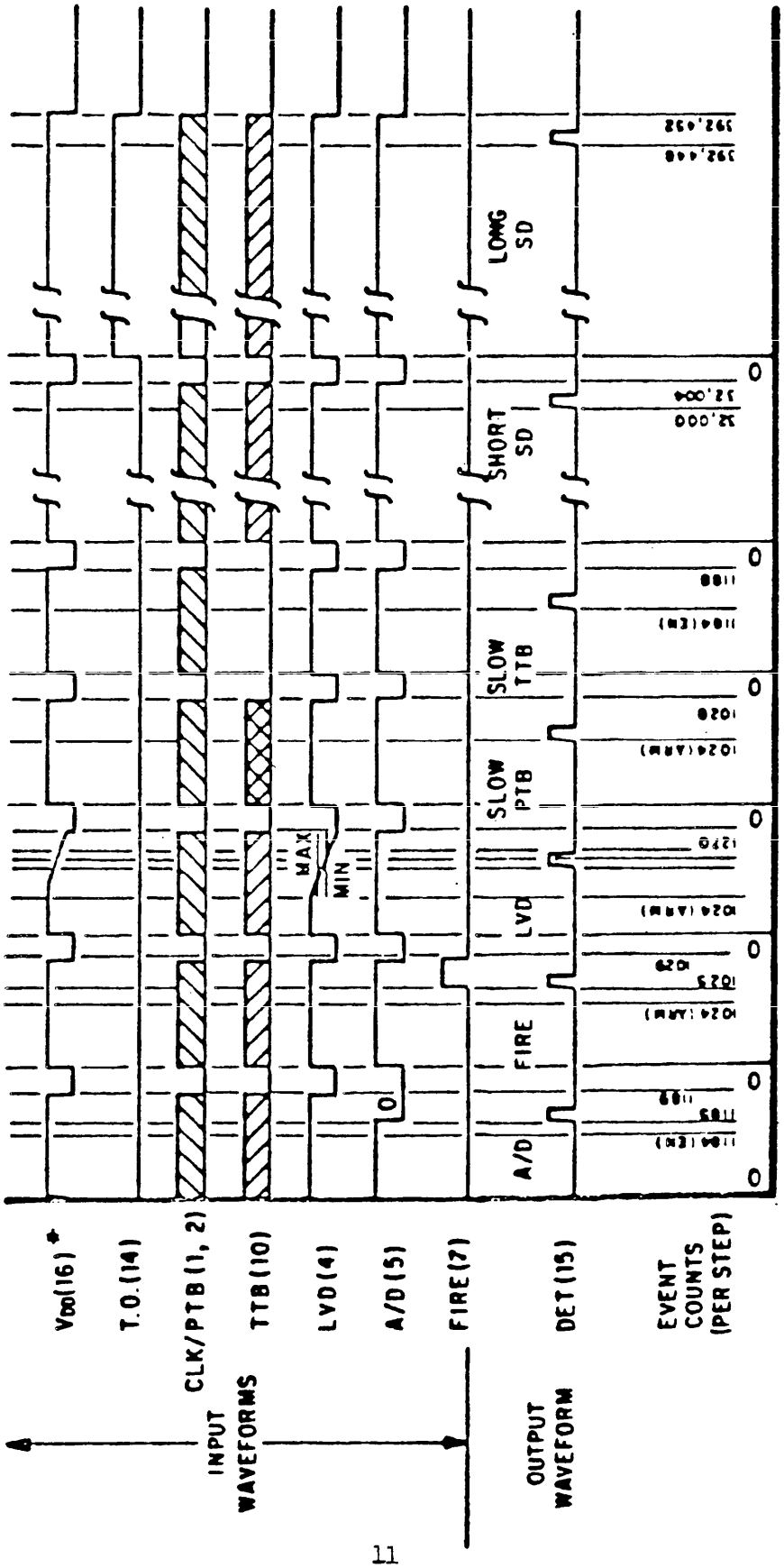


FIGURE 3 Input/Output Waveform

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3.4.9.2 Notes on figures 3 and 4. All pulse magnitudes are equal to V_B except on Pin 4 as shown in Figure 3 and explained in Para. 3.3.9.3 below.

All pulse counts are referred to the zero start time. One pulse count is equal to one pulse of the CLK/PTB (Pins 1-2) as explained in Para. 3.4.5.

Count CLK periods (T) during PTB pulse duration (64T) for count 1024. Count PTB periods (384T) for count 1024.

Symbol Terminology is as follows:



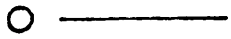
Repetitive Pulses at CLK/PTB Frequency (F_P).



Repetitive Pulses at TTB Frequency (F_T). Where: $3/5\ F_P \leq F_T \leq 4/5\ F_P$.



Repetitive Pulses at a Frequency of $2XF_P$.



Removal of a High Signal; not an applied low signal.



Incorporate Requirements of Para. 3.3.9.4 on Convenient Tests in this sequence. The test of 3.3.9.4 may be done as a separate test if desired. Ground all I.C. pins when $V_B = 0$ (between tests).

3.4.9.3 Input terminal 4. The input voltage levels applied to the input terminal 4 during the sequence test must be the maximum and minimum LVD voltages specified in Table II (Para 3.4.2) for all values of supply voltages (V_B).

3.4.9.4 V_B and pin 4 rise time. Pin 4 shall be connected to V_B as shown in Figure 4.

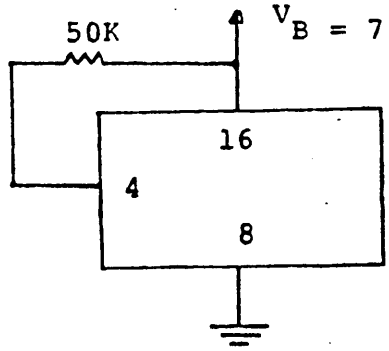


FIGURE 4

V_B rise time shall be set to 1.0 MSEC and 500.0 MSEC in successive tests. If this test is done as a separate test, tie terminals 2 and 10 to ground during the test.

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3.5 Performance.

3.5.1 High temperature storage. I.C.'s shall be subjected to testing in accordance with MIL-STD-883, Method 1008, Test Condition C, shall meet the final electrical requirements as specified in paragraph 3.5.15. There shall be no evidence of visual damage due to temperature exposure.

3.5.2 Temperature cycling. I.C.'s when tested as specified in MIL-STD-883, Method 1010, Test Condition C, shall meet the final electrical requirements listed in the paragraph 3.5.15.

3.5.3 Solderability. External leads, including tapered portion, must be electro tin plated or soldered dipped per MIL-P-38510, paragraph 3.5.6.2 (a) or (b) and if electro tin plated, the minimum plating must be 150 microinches. Parts must be solderable per MIL-STD-883, Method 2003.

3.5.4 Seal.

3.5.4.1 Fluorocarbon gross leak. I.C.'s shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition C.

3.5.4.2 Tracer gas (HE) fine leak. I.C.'s shall meet the requirements specified in MIL-STD-883, Method 1014, Test Condition A or B.

3.5.5 Thermal shock. I.C.'s, when subjected to the shock test specified in MIL-STD-883, Method 1011, Test Condition C, 15 cycles shall meet the final electrical requirements as specified in para 3.5.15 after exposure.

3.5.6 Constant acceleration. I.C.'s, when subjected to the shock test specified in MIL-STD-883, Method 2002, Test Condition E, Y_1 axis only, shall meet final electrical requirements as specified in paragraph 3.5.15.

3.5.7 Bond strength. Wires from the circuit die to the package header shall be aluminum with minimum bond strength equal to 3.0 gram pre-seal force. Bond strength shall be sufficient to meet the Class B requirements of MIL-STD-883, 2011.

3.5.8 Resistance to solvent. The marking of I.C.'s shall remain legible when subjected to resistance to solvent testing of MIL-STD-883, Method 2015, except paragraph 2.1.d.

3.5.9 Physical dimensions. I.C.'s shall comply with requirements specified in MIL-STD-883, Method 2016.

3.5.10 Internal visual. I.C.'s shall comply with requirements of MIL-STD-883, Method 2010, Condition B.

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documents not listed in the POD132 shall be the issue of the nongovernment docu-

3.5.11 External visual. The finished component shall meet the requirements of MIL-STD-883, Method 2009, there shall be no evidence of nonconformance with the detailed drawings, the absence of any required feature, damage, corrosion or contamination.

3.5.12 Lid torque. I.C.'s, when tested as specified in MIL-STD-883, Method 2024, shall exhibit no evidence of breakage, loosening, separation, or relative motion between the lid and the base when examined at a magnification between 10X and 20X.

3.5.13 First article. This specification makes provisions for first article inspection. Requirements for submission of first article samples by the contractor shall be as specified in the contract.

3.5.14 Workmanship. All I.C.'s shall be manufactured, processed and tested in a careful and workmanship manner in accordance with good engineering practice and in accordance with the requirements of the specification.

3.5.15 End point electrical. All I.C.'s shall meet the electrical requirements of Tables II, and III of this specification. The component input to output coding shall be as described in the test pattern shown in Table III. The pattern in Table III shall be run with VDD values of 5, 6, and 7 volts with input waveforms as shown in Figure 3. The pattern should be repeated at the 3 VDD values with all output pins open (unloaded) to meet the dynamic IDD requirements of Table II. In addition, Pin 3 (PTB) should provide a divide by 384 function (16 2/3% duty cycle) of the input at Pin 7. This output to be checked at any time during the test.

3.5.16 Pre-seal bake. Eutectic attached substrate and sealing glass shall be baked at 125°C in a dry nitrogen atmosphere for a specified interval (16 hours minimum for eutectic attached substrate and 2 hours minimum for sealing glass) immediately prior to sealing.

3.5.17 Screening. Prior to final inspection, the finished component shall be screened in accordance with paragraph 4.6. Components subjected to the screening procedures shall be capable of functioning and meeting the requirements of 3.4.

3.5.18 Mechanical shock. The I.C.'s, when loaded in a 155MM test vehicle projectile, shall be capable of surviving the mechanical shock requirement as specified in drawing 9317646.

3.5.19 Country of manufacturer. The component specified herein shall be manufactured, assembled and tested within the United States or its territories or by a foreign source complying with the provisions of the design standardization manual, DOD 4120.3-M. para. 4-106.6, subject to the approval of the Government Technical Agency (See 6.4).

4. QUALITY ASSURANCE PROVISIONS

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TABLE III. QUALIFICATION TEST

SYMBOL	EXAMINATION OR TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED	NO. OF DEFECTS ALLOWED
	GROUP 1-A In Process Screening		3.5.18	4.6				100%	
	GROUP 1-B Physical Dimension Marking/Resistance to Solvent External Visual Internal Visual Bond Strength Lid Torque Thermal Shock Solderability		 3.5.9 3.5.8 3.5.11 3.5.10 3.5.7 3.5.12 3.5.5 3.5.3	MIL-STD-883 2016 2015 except para 2.1.d 2009 2010 2011 2024 1011 1010				MIL-STD-883 Method 5005 Class B Sample Plan	
	GROUP II (End Point Electrical Tests)							LTPD 3	
I_L	Leakage Current	V_B = 7.5V	3.5.15	4.5.2		5.0	uA		
I_{OSC}	Dynamic Load (Oscillator) Current	V_B = 7.5V	3.5.15	4.5.3		95.0	uA		
	Sequential Operation	V_B = 5.0	3.5.15	4.5.5	(Sec. 3.4.9)				
	LVD Function Voltage	(see 4.5.6)	3.5.15	4.5.6	1.7	3.2			

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TABLE III. QUALIFICATION TEST (Continued)

SYMBOL	EXAMINATION OR TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED	NO. OF DEFECTS ALLOWED
	Input Switching Voltage	V_B = 7.5V	3.5.15	4.5.7	0.7	5.4	V	LTPD 3	
	V_B and Pin 4 rise Time During Sequential Operation	V_B = 7.5V	3.4.9.4	4.5.5	3.4.9.4				
	"1" Output Voltage Terminal 15	V_B = 5.0V	3.5.15	4.5.4.1	2.5	-	V		
	"0" Output Voltage Terminal 15	V_B = 5.0V	3.5.15	4.5.4.2	-	2.5	V		
	Input Resistance Terminal 5 & 7	V_B = 5.0V	3.5.15	4.5.8	2.0	-	V		
	"1" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.9	2.5	-	V		
	"0" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.10	-	2.5	V		

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TABLE III. QUALIFICATION TEST (Continued)

SYMBOL	EXAMINATION OR TEST	CONDITIONS	REQ. PARA	METHOD PARA	LIMITS			NO. OF SAMPLES TO BE INSPECTED	NO. OF DEFECTS ALLOWED
					MIN	MAX	UNITS		
	GROUP III-A (TA = + 63°C End Point Electronic Tests)								
I_L	Leakage Current	V_B = 7.5V	3.5.15	4.5.2	-	32	uA	LTPD 5	
I_{OSC}	Dynamic Load (Oscillator Current)	V_B = 7.5V	3.5.15	4.5.3	-	120	uA		
	"1" Output Voltage Terminal 15	V_B = 5.0	3.5.15	4.5.4.1	25	-	V		
	"0" Output Voltage Terminal 15	V_B = 5.0	3.5.15	4.5.4.2	-	25	V		
	"1" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.4.2	-	2.5	V		
	"0" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.9	-	2.5	V		
	Sequential Operation	V_B = 5.0	3.5.15	4.5.5	(See 3.4.9)				
	LVD Function Voltage	(See 4.5.6)	3.5.15	4.5.6	1.5	3.5	V		

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TABLE III. QUALIFICATION TEST (Continued)

SYMBOL	EXAMINATION TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED	NO. OF DEFECTS ALLOWED
	GROUP III - B (T_A = -32°C End Point Electronic Tests)								
I_L	Leakage Current	V_B = 7.5V	3.5.15	4.5.2	-	32	uA	LTPD 5	
I_{OSC}	Dynamic Load (Oscillator) Current	V_B = 7.5V	3.5.15	4.5.3	-	120	uA		
	"1" Output Voltage Terminal 15	V_B = 5.0	3.5.15	4.5.4.1	2.5	-	V		
	"0" Output Voltage Terminal 15	V_B = 5.0	3.5.15	4.5.4.2	-	2.5	V		
	"1" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.9	2.5	-	V		
	"0" Output Voltage Terminal 11	V_B = 5.0	3.5.15	4.5.9	-	2.5	V		
	Sequential Operation	V_B = 5.0	3.5.15	4.5.5	(See 3.4.9)				
	LVD Function Voltage	(See 4.5.6)	3.5.16	4.5.6	1.5	3.5	V		
	GROUP IV Mechanical Shock	(See Dwg. 9317646)	3.5.18	(See Dwg. 9317646)	-	-	-		

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4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First article inspection.

b. Quality conformance inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2.

4.3.2 Inspection to be performed. The first article parts submitted in accordance with Paragraph 4.3.1 shall be subjected by the Government to any or all the requirements of the applicable drawings and to the inspection of Table V. All defects are classified as Major unless otherwise specified.

4.3.3 Inspection conditions. Unless otherwise specified, all examinations and tests shall be made under the following conditions:

a.	Temperature	25 ± Deg C
b.	Humidity	10 to 75% RH
c.	Pressure	29 ± inches of Hg
d.	Voltage and current measure	± 2%

4.3.4 Rejection. See MIL-A-48078. Resubmission for the first article tests shall be in accordance with the appropriate contract clause.

4.3.5 Re-qualification. I.C.'s shall be subject to another complete qualification if any changes are made which require an interfix number change (see paragraph 5.3).

4.4 Quality conformance inspection. The quality conformance inspection listed in paragraph Table VI shall be performed on each inspection lot submitted for acceptance.

4.4.1 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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TABLE IV. QUALITY CONFORMANCE TEST

SYMBOL	EXAMINATION TEST	CONDITIONS	REQ PARA	METHOD PARA	LIMITS MIN	LIMITS MAX	LIMITS UNITS	NO. OF SAMPLES TO BE INSPECTED	NO. OF DEFECTS ALLOWED
	GROUP 1 In Process Screening		3.5.18	4.6				100%	
	Physical Dimensions		3.5.9	2016				MIL-STD-883 Method 5005	
	Marking/Resistance to Solvent		3.5.8	2015 except para 2.1.d					
	External Visual		3.5.11	2009				Class B Sample Plan	
	Bond Strength		3.5.7	2011					
	Lid Torque		3.5.12	2024					
	Thermal Shock		3.5.5	1011					
	Solderability		3.5.3	1010					
	GROUP II (Same as Group II, Table III)		3.5.15					LTPD 5	
	Sequential Operation	V_B = 5.5V T_A = -32°C	3.5.15	4.5.5	(See 3.4.9)			LTPD 10	
		V_B = 7.2V T_A = -32°C							
		V_B = 5.5V T_A = +63°C							
		V_B = 7.2V T_A = +63°C							

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4.4.2 Rejection. If any unit under test fails to comply with any of the applicable requirements, the unit shall be rejected. Lots shall not be resubmitted for these tests without prior approval of the procuring activity. Acceptance of product shall be discontinued pending investigation of the failure and approval by the design activity of corrective action proposed by the contractor.

4.5 Electrical tests.

4.5.1 Test method. The test methods which follow are intended to convey the idea of what is to be tested and provide a simple bench method of doing it. It is recognized that automatic (computerized) tests will differ as to implementation of the test.

4.5.2 Leakage current. The microcircuit shall be connected as shown in Figure 5. The supply voltage shall be applied between terminals 16 (V_B) and 8 (ground). A minimum of 50 milliseconds shall elapse before measuring the leakage current. After measuring the current with circuit configured as shown in Figure 5, terminal 2 and 10 shall be switched from ground to V_B (see Figure 6). The leakage current with the circuit configured as shown in Figure 6 shall be measured. Leakage current shall be as specified in Table II.

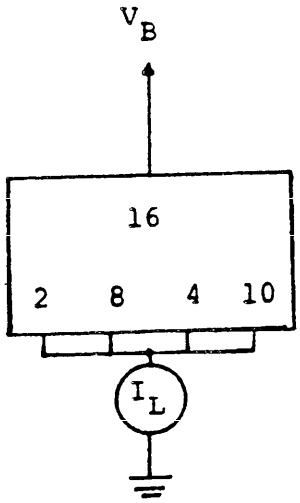


FIGURE 5

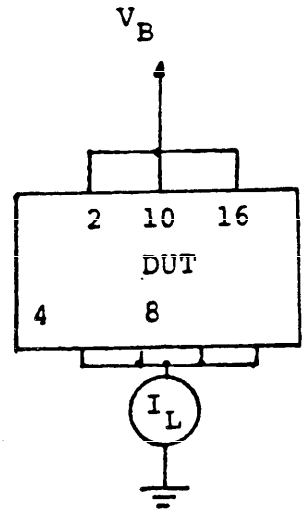


FIGURE 6

4.5.3 Dynamic load (oscillator) current. The component is connected as shown in Figure 1B. It shall operate with load current within the limits of Table II. Current shall be observed after arming occurs.

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4.5.4 Output voltage.

4.5.4.1 With the component connected as shown in Figure 7, pulses as defined in para. 3.4.5 are applied to terminal 2. The "1" output voltage is measured at terminal 15 after 32,000 pulses.

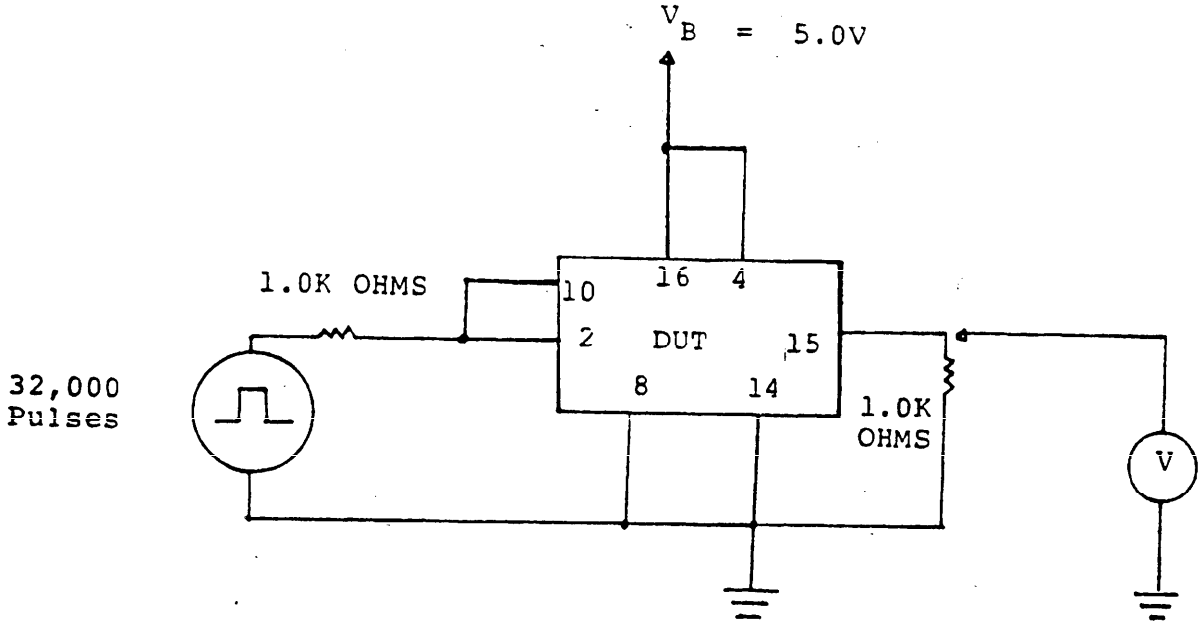


FIGURE 7 "1" OUTPUT VOLTAGE TEST

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4.5.4.2 With the component connected as shown in Figure 8, any number of pulses as defined in para. 3.4.5 are applied to terminal 2 as necessary to obtain a "0" output voltage on terminal 15.

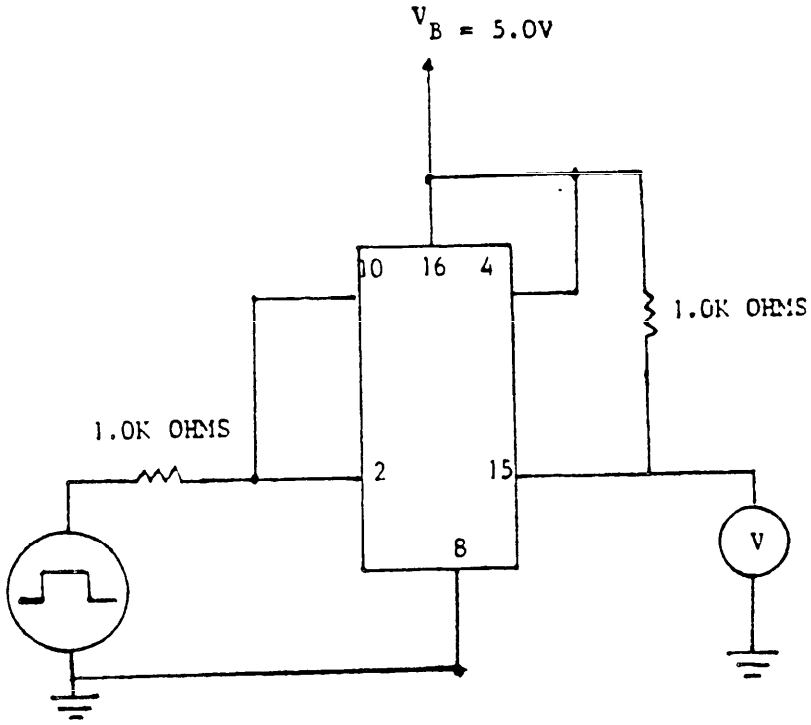


FIGURE 8 "0" OUTPUT VOLTAGE TEST

4.5.5 Sequential operation. The component is connected as shown in Figure 2 and the waveforms shown in Figure 3 are applied. Operation shall be as described by the output waveforms in Figure 3 and paragraph 3.4.9.

4.5.6 LVD function voltage terminal 15. With the circuit connected as shown in Figure 9, and 1024 pulses input on Pin 2, the voltage on Pin 4 and 16 is decreased until a high output is observed on Pin 15. The voltage on Pin 4 at which the output on Pin 15 occurs shall be recorded as the LVD voltage.

4.5.7 Input switching threshold voltage terminal 10. With the circuit connected as shown in Figure 10, a slowly increasing voltage is applied to terminal 10. The level on Pin 11 shall be observed and when Pin 11 goes from a high to a low level, the voltage level on terminal 10 shall be measured as the threshold voltage on Pin 10.

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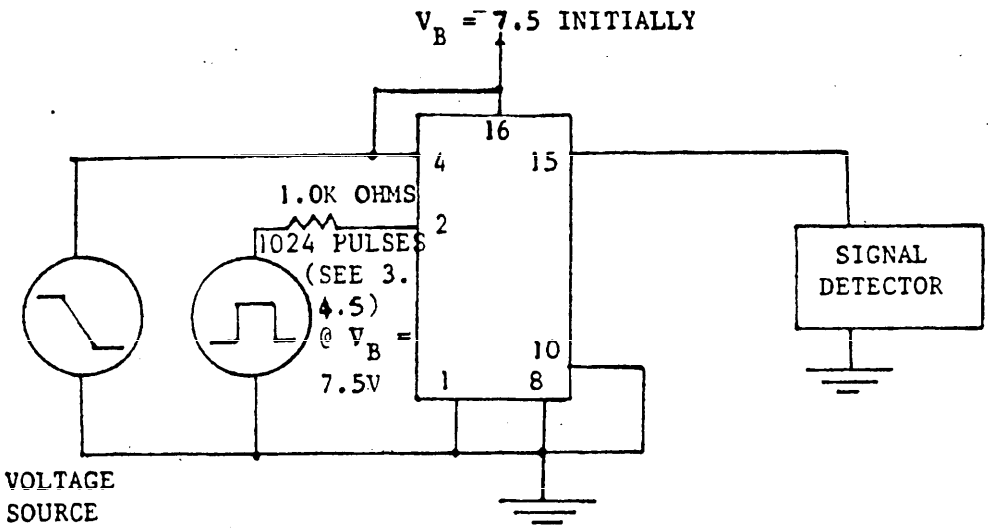


FIGURE 9 LVD VOLTAGE TEST

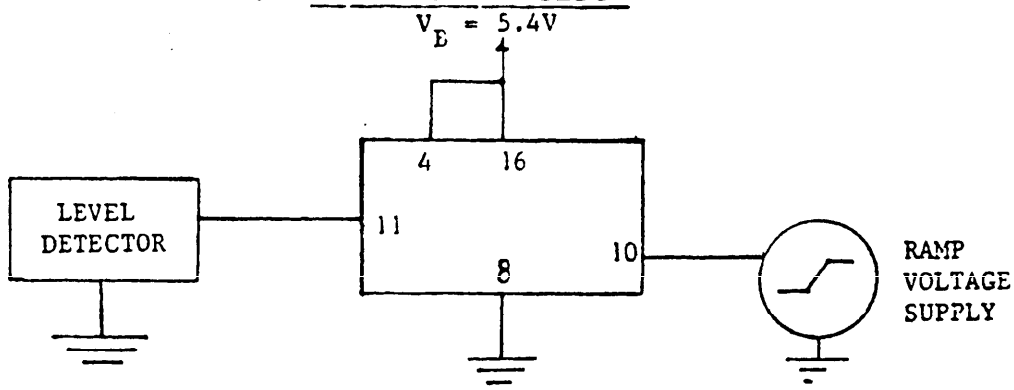


FIGURE 10 INPUT SWITCHING THRESHOLD VOLTAGE TEST

4.5.8 Input resistance. The input resistance on terminals 1, 2, 5, and 7 shall be measured by connecting the device as shown in Figure 11 and measuring the voltages on terminals 1, 2, 5, and 7. The voltages shall be as specified in Table II.

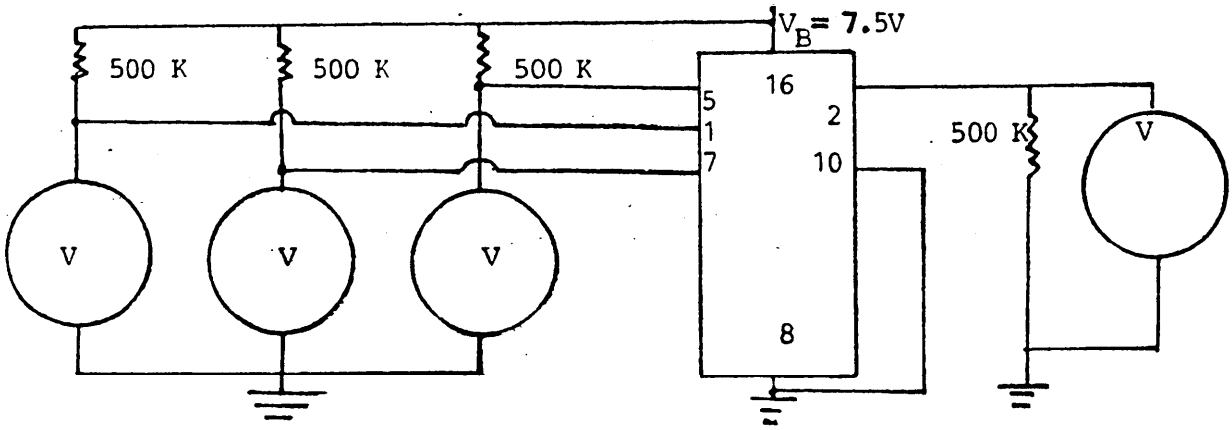


FIGURE 11 INPUT RESISTANCE TEST

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4.5.9 "1" Output voltage, terminal 11. The output voltage on terminal 11 shall be measured by connecting the device as shown in Figure 12. Open SW_1, measure output on terminal 11.

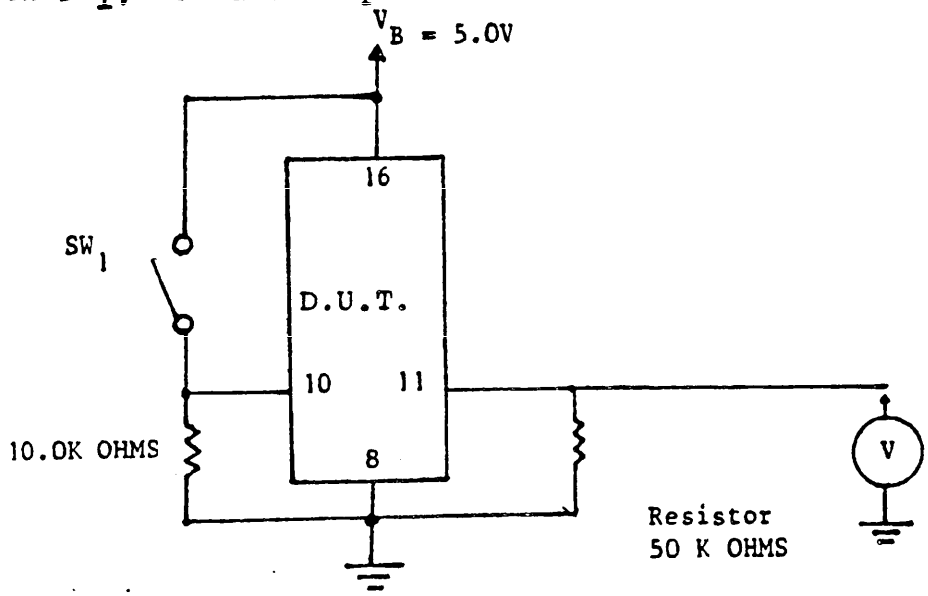


FIGURE 12 "1" OUTPUT VOLTAGE TEST, TERMINAL 11

4.5.10 "0" Output voltage terminal 11. The output voltage on terminal 11 shall be measured by connecting the device as shown in Figure 13.

Close SW_1, measure output on terminal 11.

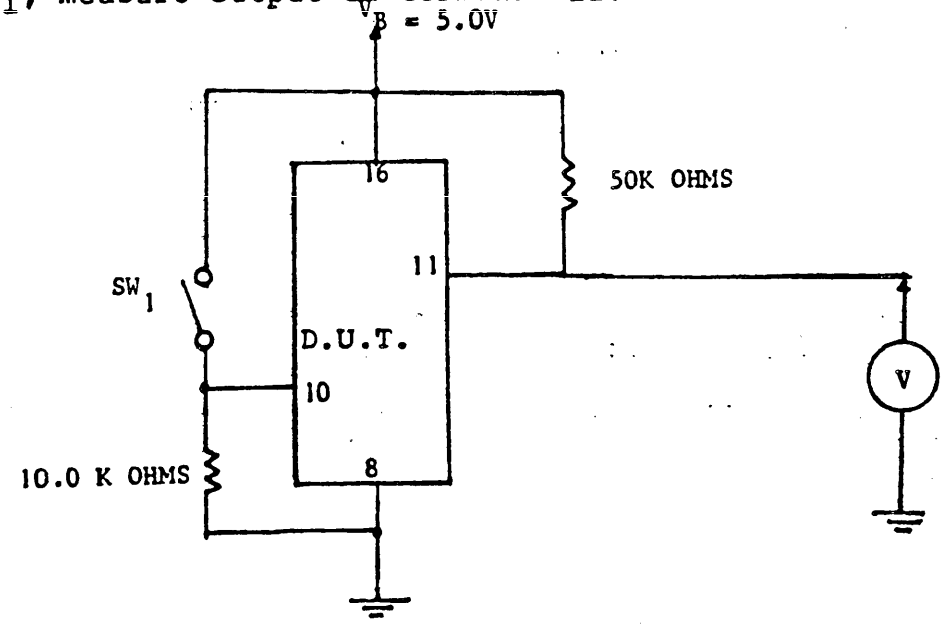
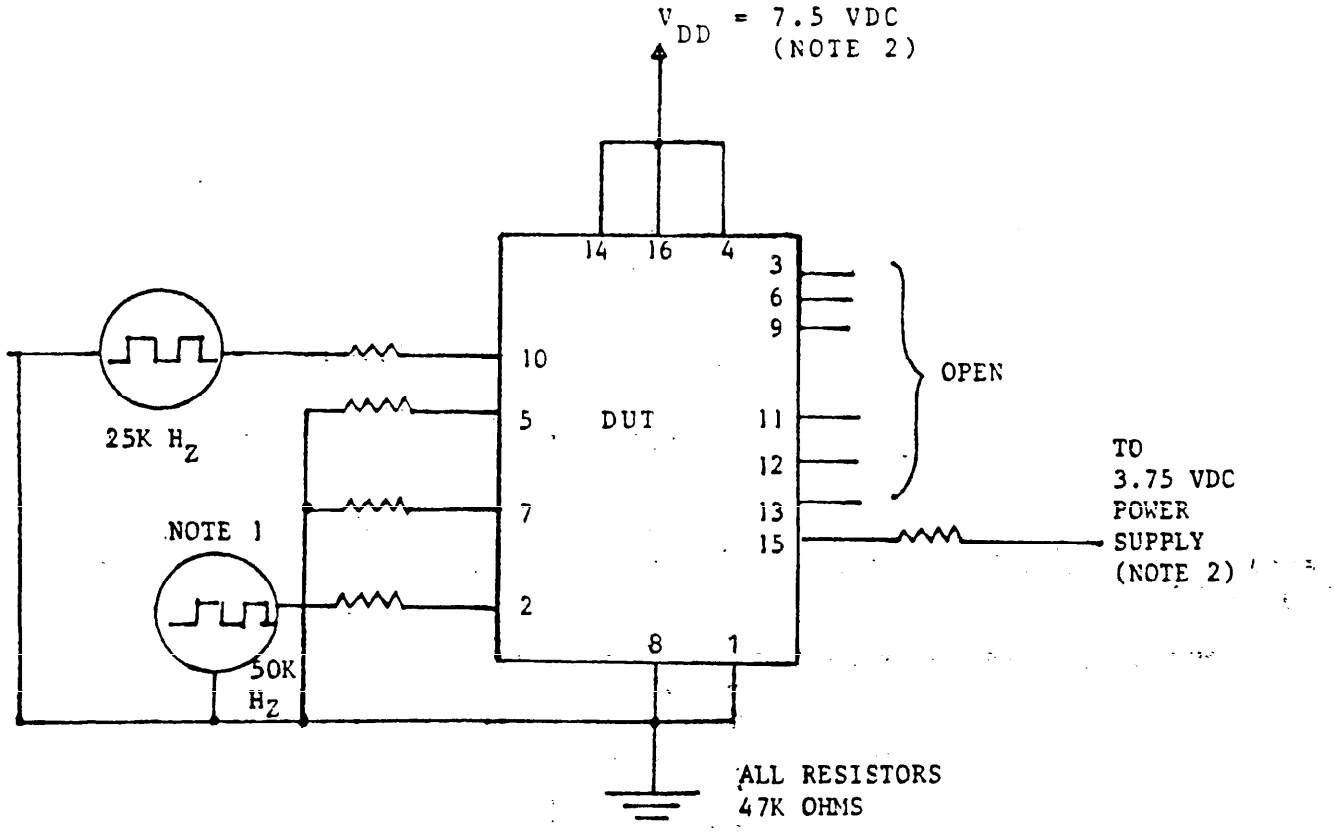


FIGURE 13 "0" OUTPUT VOLTAGE TEST, TERMINAL 11.

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4.5.11 Dynamic burn-in test. A dynamic burn-in test shall be performed at a temperature of 125°C with the component connected as shown in Figure 16. When the test is completed, the component shall be cooled to a temperature of 25°C. Then the tests in Table III Group II shall be performed.



NOTES:

1. Rack frequency 30 - 70% duty cycle.
   Amplitude of square wave: 0 and 7.5V

2. Both power supplies shall be turned off momentarily at the end of each hour of operation to allow all logic circuits to return to the reset state.

FIGURE 14 DYNAMIC BURN-IN TEST

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4.5.12 Capacitance - Coupled feedback test. The internal capacitance coupling between the output at terminal 9 and the input at terminal 10 shall be tested by measuring the amplitude of the spike signal observed at terminal 10 with device connected as shown in Figure 15. The signal shall be observed using a storage oscilloscope (Tetetronix 7633 with vertical amplifier 7A26 and time base 7B53A, on an equivalent instrument). The amplitude of the spike signal (V_{FB}) shall be measured as shown in Figure 15. The amplitude shall be as specified in Table 2.

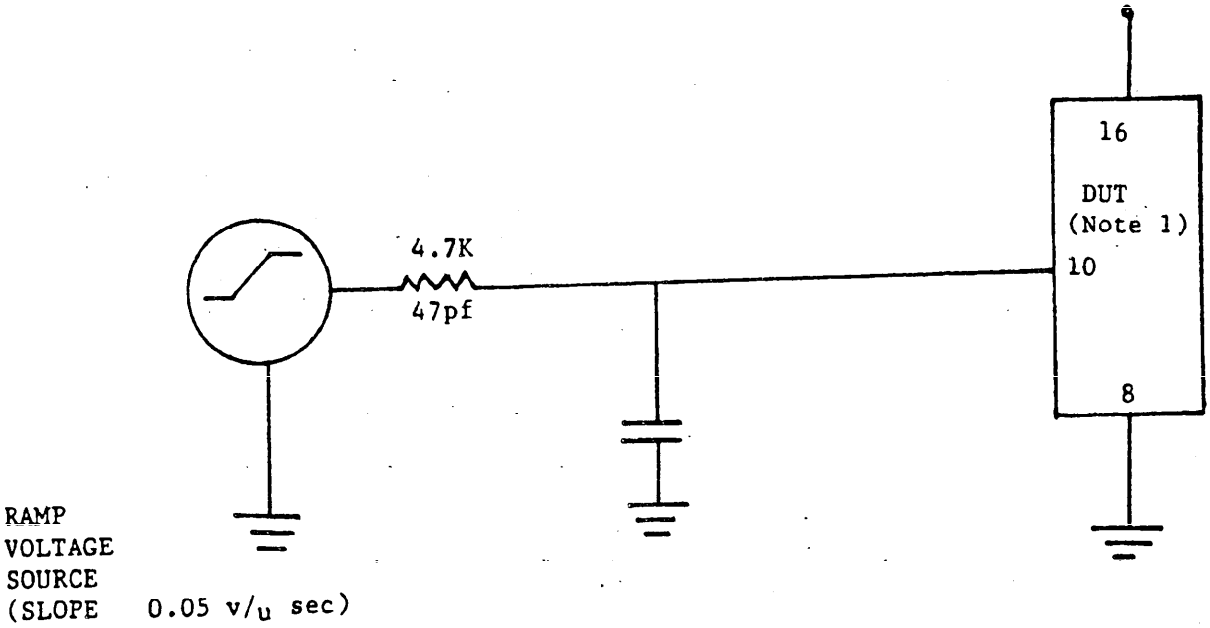
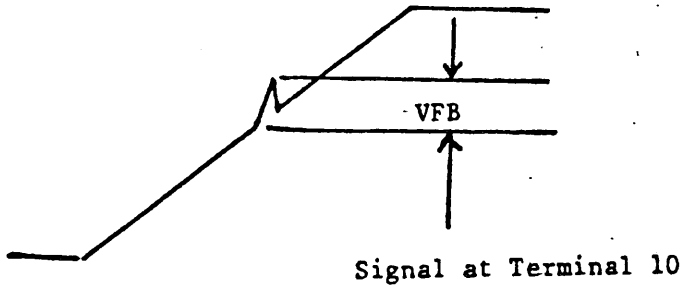


FIGURE X. FEED-BACK SIGNAL TEST

NOTE 1.



Signal at Terminal 10

FIGURE 15. FEEDBACK SIGNAL TEST

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4.6 Screening. Screening shall be in accordance with MIL-STD-883, method 5004 with the exception of the following:

Screen	Method
Burn-In test	1015 at 125°C for 160 hours (See paragraph 4.5.11)
Post Burn-in	1015 at 125°C ALL ELECTRICAL Parameters per TABLE I.

5.

5.1 Packaging. Circuit shall be packaged with a conductive material so that damage from static charge or stray voltage is prevented.

5.2 Marking. Each circuit shall be marked with the last three digits of the part number and revision letter. (Example: 165A) The circuit shall also be marked with a lot number traceable to the vendor's manufacturing records. The vendor may add additional markings for his use provided the additional markings do not interfere with the required markings.

5.3 Inspection lot formation (Lot Identification). To meet the requirements for material traceability required by contract, each integrated circuit lot will contain only one interfix number.

The lot identification/date number shall be preceded by a three-digit interfix number. The interfix number shall be an integral part of the lot identification number, and must be used to identify those lots produced homogeneously in one continuous process. Any of the conditions described below shall be cause for changing the interfix number.

5.3.1 Change in manufacturing process. Changes representative of those which are subject to this requirement are any basic design changes in the manufacturing process such as:

a. Die structure topography changes (e.g., double-diffused, epitaxial, isolation).
b. Mask changes or redesign which alter die size.

c. Change in passivation or classification material thickness or technique (including addition or deletion of passivation).

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d. Change in die substrate or element attach material, method or location.

e. Changes in wire dimensions.

f. Package or lid structure and material changes or changes to the internal cavity geometry.

5.3.2 Change in manufacturing location. If the active line moves from one geographical location to another, the interfix number will be changed.

5.3.3 Interruption in manufacturing process over 90 days. A complete line shutdown of more than 90 days duration will be cause for a change of the interfix number.

6. NOTES

6.1 Ordering data. See MIL-A-48078 (PA)

6.2 Submission of inspection equipment designs for approval. (See MIL-A-48078). Submit equipment designs as required to Commander: USAMCCOM, ATTN: AMSMC-QAT-I(D), Dover, NJ 07801-5001. Request that the letter of submittal state contractor, contract number, specification number, item nomenclature and classification of defects or test paragraph.

6.3 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as, US Army Armament Research and Development Command, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.4 Distribution of ammunition data cards, inspection and test results. Distribution of ammunition data cards, inspection, and test results shall include Commander: AMCCOM, Dover, NJ 07801-5001, ATTN: AMSMC-QAT-M(D).

6.5 Subject term (key word) listing.

Dynamic burn-in test
Dynamic load, Oscillator current
Integrated Circuit, Digital, CMOS Contral and Timing Base
Leakage Current test
LVD Functioning
Mechanical Shock
Sequential Operation Test
Stress Screening

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A300)

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MIL-I-48632(AR)	INTERGRATED CIRCUIT, DIGITAL, CMOS CONTROL AND TIMING BASE, MONOLITHIC, SILICON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
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